

IT Solera PVT LTD

AI & Data Science– Advanced Internship Tasks

Due Date: 20 February 2026

Overview

As part of your AI & Data Science **Internship** at **IT SOLERA PVT LTD** , you are required to complete **at least 4 out of the 5 advanced tasks** listed below within the given timeline.

These tasks are designed to provide you with practical exposure to **real-world data science problems**, including **classification, clustering, time series forecasting, explainable AI (XAI), and business intelligence**.

You are encouraged to complete all tasks for deeper learning and a stronger portfolio. You'll work with popular datasets and industry-standard tools and libraries such as `pandas`, `numpy`, `scikit-learn`, `xgboost`, `shap`, `matplotlib`, `seaborn`, `Prophet`, and `Streamlit`.

Advanced Task Set

Task 1: Term Deposit Subscription Prediction (Bank Marketing)

Objective:

Predict whether a bank customer will subscribe to a term deposit as a result of a marketing campaign.

Dataset:

Bank Marketing Dataset (UCI Machine Learning Repository)

Instructions:

- Load and explore the dataset
- Encode all categorical features properly

- Train classification models (e.g., Logistic Regression, Random Forest)
- Evaluate the models using Confusion Matrix, F1-Score, and ROC Curve
- Use SHAP or LIME to explain at least 5 model predictions

Skills Gained:

- Classification modeling
- Feature encoding
- Model interpretability (Explainable AI - XAI)
- Customer behavior analysis

Task 2: Customer Segmentation Using Unsupervised Learning

Objective:

Cluster customers based on spending habits and propose marketing strategies tailored to each segment.

Dataset:

Mall Customers Dataset

Instructions:

- Conduct Exploratory Data Analysis (EDA)
- Apply K-Means Clustering to segment customers
- Use PCA or t-SNE to visualize the clusters
- Suggest relevant marketing strategies for each identified segment

Skills Gained:

- Unsupervised learning (K-Means)

- Dimensionality reduction (PCA, t-SNE)
- Customer segmentation
- Strategy development based on data insights

Task 3: Energy Consumption Time Series Forecasting

Objective:

Forecast short-term household energy usage using historical time-based patterns.

Dataset:

Household Power Consumption Dataset

Instructions:

- Parse and resample the time series data
- Engineer time-based features (e.g., hour of day, weekday/weekend)
- Compare performance of ARIMA, Prophet, and XGBoost models
- Plot actual vs. forecasted energy usage for visualization

Skills Gained:

- Time series forecasting
- Feature engineering
- Model comparison and evaluation (MAE, RMSE)
- Temporal data visualization

Task 4: Loan Default Risk with Business Cost Optimization

Objective:

Predict the likelihood of a loan default and optimize the decision threshold based on cost-benefit analysis.

Dataset:

Home Credit Default Risk Dataset

Instructions:

- Clean and preprocess the dataset
- Train binary classification models (e.g., Logistic Regression, CatBoost)
- Define business cost values for false positives and false negatives •

Adjust the model threshold to minimize total business cost

Skills Gained:

- Binary classification modeling
- Cost-based evaluation metrics
- Risk modeling and scoring
- Feature importance analysis

Task 5: Interactive Business Dashboard in Streamlit**Objective:**

Develop an interactive dashboard for analyzing sales, profit, and segment-wise performance.

Dataset:

Global Superstore Dataset

Instructions:

- Clean and prepare the dataset
- Build a Streamlit dashboard with filters (Region, Category, Sub-Category)
- Display key performance indicators (KPIs) using charts:
 - Total Sales
 - Profit
 - Top 5 Customers by Sales

Skills Gained:

- Business Intelligence (BI) dashboarding
- Data storytelling
- User interactivity with Streamlit
- Visual KPI analysis

Submission Requirements (Per Task)

Each completed task must be **uploaded to your GitHub repository** and shared via **Google Classroom**.

Checklist for Each Task:**1. Jupyter Notebook**

- Problem Statement and Objective
- Dataset description and loading
- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Model building and evaluation
- Visualizations (charts, plots, graphs)
- Final conclusion with insights

2. Code Quality

- Well-structured, readable, and commented code

3. GitHub Repository

- Clear and descriptive repository name
- README.md file including:

- Task objective
- Your approach
- Results and findings

4. Submission on Google Classroom

- Share the link to your GitHub repository after completing each task

Important Notes

- **Complete at least 4 out of the 5 tasks by 20 February 2026**
- You are encouraged to complete all 5 tasks
- Reach out to mentors if you need help or feedback
- Showcase your work on LinkedIn or GitHub for better visibility

Task Presentation Requirement

After completing each task, interns are required to present their work.

- Short presentation (10-15 minutes)
- Explain problem statement, approach, key visualizations, results, and insights
- Presentations will be conducted online
- Interns must be able to clearly explain their own code

Note:

- Tasks without presentation will be considered incomplete.
- Missing even one presentation will result in immediate removal from the internship program.